Schrödinger Bridge in 1-D via Staggered-Grid ADMM

Motivation · Formulation · Discretisation · Projection Solvers · Stability

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The Two Approaches to Regularised OT

Same underlying problem: move mass from ρ_0 to ρ_1 at minimum cost, with entropic regularisation $\varepsilon > 0$.

Static / Sinkhorn approach

Solve the $N_x \times N_x$ coupling problem:

$$\min_{\pi \geq 0} \langle C, \pi \rangle + \varepsilon \langle \pi, \log \pi \rangle, \quad \pi \mathbf{1} = \mathbf{a}, \ \pi^\top \mathbf{1} = \mathbf{b}.$$

Solution: $\pi^* = \mathrm{diag}(\mathbf{u})\, K\, \mathrm{diag}(\mathbf{v})$,

$$K_{ij} = e^{-C_{ij}/\varepsilon}, \quad C_{ij} = |x_i - y_j|^2.$$

Sinkhorn iterations: alternating row/column scaling of K.

Dynamic / Benamôu–Brenier approach

Solve the space-time PDE-constrained problem on an $N_x \times N_t$ grid:

$$\min_{\rho, m} \ \frac{1}{2} \iint \frac{m^2}{\rho} \quad \text{s.t.} \quad \partial_t \rho + \nabla_x \cdot m = \varepsilon\, \Delta_x \rho.$$

ε appears as a *PDE diffusion coefficient*, never inside an exponential.

Equivalent as $\varepsilon \to 0$

Both formulations share the same ε: as $\varepsilon \to 0$ they recover the same unregularised W_2 transport. The key question is what happens *numerically* when ε is small.

Sinkhorn Breaks Down for Small ε

Problem 1: Numerical underflow in K

$K_{ij} = e^{-C_{ij}/\varepsilon}$ underflows to `0.0` when $C_{ij}/\varepsilon \gtrsim 700$ (IEEE 754 double).

Example: $C_{ij} = |x_i - y_j|^2 \sim 0.1$, $\varepsilon = 10^{-4}$: $K_{ij} = e^{-1000} \approx 0$.

Log-domain Sinkhorn helps but adds complexity and does not cure the ill-conditioning.

Problem 2: Exponential iteration count

Hilbert metric analysis: convergence factor $r \approx 1 - 2e^{-\Delta c/\varepsilon} \to 1$ as $\varepsilon \to 0$.

Iterations needed: $O(e^{\Delta c/\varepsilon})$ — **exponential**.

For $\Delta c \sim 1$, $\varepsilon = 10^{-3}$: $\sim e^{1000}$ iterations.

Problem 3: Dense N_x^2 matrix

K is $N_x \times N_x$ and dense. Each Sinkhorn iteration costs $O(N_x^2)$. Total cost $O(N_x^2/\varepsilon)$.

No spatial sparsity is exploited.

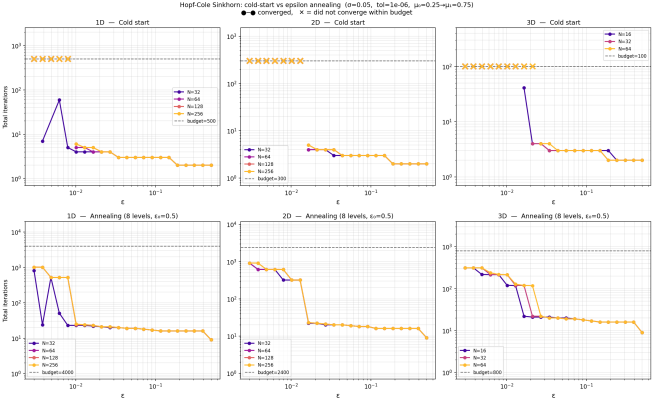
Dynamic formulation: no exponentials

ε enters the FP equation as a coefficient; numerics are independent of $e^{-1/\varepsilon}$.

ADMM projection costs $O(N_x N_t \log N_x)$ per iteration (using DCT and tridiagonal solves).

The effective N is the *space-time grid size* $N_x N_t$, not the quadratic coupling matrix N_x^2.

Sinkhorn Iteration Count Explodes as $\varepsilon \to 0$



Hopf-Cole Sinkhorn (1-D/2-D/3-D). **Top**: cold start; **Bottom**: ε-annealing (8 levels). $\times$ = did not converge within budget (500/4000 iters). Iteration count grows *exponentially* as $\varepsilon \to 0$; grid size and annealing only shift the curve. The dynamic BB/ADMM formulation avoids this entirely.

Benamôu–Brenier and Schrödinger Bridge

Dynamic OT

$$\min_{\rho,m} \int_0^1 \int_0^1 \frac{m(t,x)^2}{2\,\rho(t,x)}\,dx\,dt \quad \text{s.t.} \quad \underbrace{\partial_t \rho + \nabla_x \cdot m = \varepsilon\,\Delta_x \rho}_{\text{Fokker–Planck}},\ \rho(0,\cdot) = \rho_0,\ \rho(1,\cdot) = \rho_1,\ m|_{\partial\Omega} = 0.$$

- $\varepsilon = 0$: continuity equation, objective $= W_2(\rho_0, \rho_1)^2$.
- $\varepsilon > 0$: Schrödinger bridge — most likely diffusion path from ρ_0 to ρ_1 under Brownian motion with diffusivity ε.
- **Gaussian-to-Gaussian exact solution** ($\rho_0 = \mathcal{N}(\mu_0, \sigma_0^2)$, $\rho_1 = \mathcal{N}(\mu_1, \sigma_1^2)$):

$$\sigma_{\mathrm{SB}}^2(t) = (1-t)^2\sigma_0^2 + 2t(1-t)\sqrt{\varepsilon^2 + \sigma_0^2\sigma_1^2} + t^2\sigma_1^2.$$

ADMM splitting: introduce consensus copy $(\tilde{\rho}, \bar{m})$ constrained to $\mathcal{C}$:

$$\min \quad \underbrace{f(\rho, m)}_{\text{kinetic energy}} + \underbrace{\iota_{\mathcal{C}}(\tilde{\rho}, \bar{m})}_{\text{FP constraint}} \quad \text{s.t.} \quad (\rho, m) = (\tilde{\rho}, \bar{m}).$$

Optimise First or Discretise First?

Discretise first, then optimise

Discretise the FP constraint on a grid: $A_h(\rho_h, m_h) = 0$. Run ADMM in the finite-dimensional space. Projection = linear solve $(A_h A_h^\top)\phi_h = f_h$.

Appealing for standard OT ($\varepsilon = 0$): $A_m A_m^\top = -\Delta_x$, inverted by FFT. The matvec A_h is a local stencil — GPU-friendly.

Fails for SB ($\varepsilon > 0$): $A_\rho = \partial_t - \varepsilon\Delta_x$ couples space and time. $A_\rho A_\rho^\top$ involves Δ_x^2; no GPU-friendly separable inversion exists.

Optimise first, then discretise

Derive the projection formula analytically in the *continuous* setting, then discretise the result carefully.

Advantage: boundary terms and the adjoint structure are visible before any discretisation. A staggered grid then makes $A_h^\top$ the *exact* discrete adjoint of A_h.

Challenge: even with the correct structure, a naive discretisation of the resulting linear system still fails.

Our path

A discretise-first GPU approach could not be made to work for SB. We adopt the **optimise-first** approach.

The Continuous Projection Formula: $AA^*\phi = f$

Given (μ, ψ), the ADMM projection step minimises $\frac{1}{2}\|\rho - \mu\|^2 + \frac{1}{2}\|m - \psi\|^2$ subject to $A_\rho \rho + A_m m = 0$.

KKT (ϕ: Lagrange multiplier): $\tilde{\rho} = \mu - A_\rho^* \phi$, $\tilde{m} = \psi - A_m^* \phi$. Substituting into the FP constraint gives $AA^*\phi = f$, $f = A_\rho \mu + A_m \psi$, with

$$A_\rho = \partial_t - \varepsilon \Delta_x, \quad A_\rho^* = -\partial_t - \varepsilon \Delta_x, \quad A_m = \nabla_x \cdot, \quad A_m^* = -\nabla_x.$$

The $\partial_t \Delta_x$ cross-terms cancel, yielding:

$$\boxed{-\Delta_{t,x}\phi + \varepsilon^2 \Delta_x^2 \phi = f,} \qquad \Delta_{t,x} = \partial_{tt} + \Delta_x.$$

Biharmonic-type PDE: space-time Laplacian + biharmonic in x. Well-posed with Neumann BCs; for $\varepsilon = 0$ reduces to $-\Delta_{t,x}\phi = f$.

The discretisation challenge

A naive (cell-centred, non-staggered) discretisation of this PDE satisfies the FP constraint only up to $O(\varepsilon/\Delta t)$ errors. For $\varepsilon \gtrsim 10^{-4}$, $\Delta t = 1/128$: the projection is inaccurate and ADMM *diverges*. A staggered grid is required — but even then, the discrete normal equations have a non-trivial boundary structure.

Staggered Grid and Operator Notation

Variable locations:

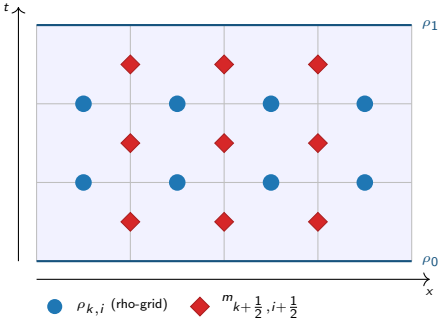
Var.	Time	Space
ρ	interior faces $(N_t-1) \times N_x$	centres
m	centres $N_t \times (N_x-1)$	interior faces

Four matrix operators:

$D_t : N_t \times (N_t-1)$	time FD (rho→phi)
$I_t^\top,\ I_t^\top : (N_t-1) \times N_t$	time avg (adjoints)
$D_x : N_x \times (N_x-1)$	space divergence

Discrete Laplacian ($N_x \times N_x$, neg. semidef.):

$$L_x = -D_x D_x^\top \ \leftrightarrow\ -\Delta_x, \quad L_x v_k = -\lambda_x(k)\, v_k.$$



DCT diagonalises L_x:

$$V L_x V^\top = -\operatorname{diag}(\lambda_x), \quad \lambda_x(k) \geq 0.$$

Same for $D_t D_t^\top$: time-Neumann Laplacian with eigenvalues $\lambda_t(l) \geq 0$.

FP Operator as a Kronecker Sum

Vectorise ϕ column-major (N_t fast) to get $\phi_{\text{vec}} \in \mathbb{R}^{N_t N_x}$. Then:

$$A = \left[\, A_\rho \;\middle|\; A_m \,\right], \qquad \text{where}$$

Density operator A_ρ

$$\boxed{A_\rho = I_{N_x} \otimes D_t \;+\; \varepsilon\, L_x \otimes I_t}$$

- $I_{N_x} \otimes D_t$: apply D_t to every spatial column.
- $\varepsilon\, L_x \otimes I_t$: average in time *then* apply L_x across space.

Momentum operator A_m

$$\boxed{A_m = D_x \otimes I_{N_t}}$$

- $D_x \otimes I_{N_t}$: apply space divergence D_x to every time row.

Adjoint $A^\top \phi_{\text{vec}} = [A_\rho^\top \phi_{\text{vec}};\; A_m^\top \phi_{\text{vec}}]$:

$$A_\rho^\top = I_{N_x} \otimes D_t^\top + \varepsilon\, L_x \otimes I_t^\top, \qquad A_m^\top = D_x^\top \otimes I_{N_t}.$$

These are used in the projection update: $(\rho_{\text{out}}, m_{\text{out}}) = (\mu, \psi) - A^\top \phi$.

Normal Equations via Kronecker Algebra

The projection solves $(AA^\top)\phi = f$. Expand using $(A \otimes B)(C \otimes D) = AC \otimes BD$:

$$\begin{aligned} A_\rho A_\rho^\top &= (I \otimes D_t)(I \otimes D_t^\top) + \varepsilon\,(L_x \otimes D_t I_t^\top + L_x \otimes I_t D_t^\top) + \varepsilon^2\, L_x^2 \otimes I_t I_t^\top \\ &= I \otimes D_t D_t^\top + \varepsilon\, L_x \otimes \underbrace{(D_t I_t^\top + I_t D_t^\top)}_{\text{rank-2!}} + \varepsilon^2\, L_x^2 \otimes I_t I_t^\top, \end{aligned}$$

$$A_m A_m^\top = (D_x \otimes I)(D_x^\top \otimes I) = \underbrace{D_x D_x^\top}_{=-L_x} \otimes I = -L_x \otimes I.$$

The rank-2 boundary identity

$$D_t I_t^\top + I_t D_t^\top = \frac{1}{\Delta t}\left(e_1 e_1^\top - e_{N_t} e_{N_t}^\top\right) \quad \text{(proved by direct computation).}$$

Substituting and collecting:

$$AA^\top = \underbrace{I \otimes D_t D_t^\top - L_x \otimes I + \varepsilon^2 L_x^2 \otimes I_t I_t^\top}_{\widetilde{P}\ (I_t I_t^\top \text{ not circulant — corrected next slide})} + \underbrace{\frac{\varepsilon}{\Delta t}\, L_x \otimes \left(e_1 e_1^\top - e_{N_t} e_{N_t}^\top\right)}_{E\ (\text{rank-2 in } t)}.$$

The $I_t I_t^\top$ Boundary Error — and Its Rank-2 Structure

$I_t I_t^\top$ has interior diagonal $\frac{1}{2}$ but **corner entries** $\frac{1}{4}$. Splitting off the circulant (periodic-BC) part $(I_t I_t^\top)_{\mathrm{circ}}$ with eigenvalues $\sigma_t(I) = \cos^2(\pi I \Delta t/2)$:

$$I_t I_t^\top = \underbrace{(I_t I_t^\top)_{\mathrm{circ}}}_{\text{DCT-diagonalisable}} - \underbrace{\tfrac{1}{4}\left(e_1 e_1^\top + e_{N_t} e_{N_t}^\top\right)}_{E_2 \text{ (rank-2!)}}.$$

So the full $AA^\top$ splits into a *circulant* base plus **two** rank-2 boundary corrections:

$$AA^\top = \underbrace{I \otimes D_t D_t^\top - L_x \otimes I + \varepsilon^2 L_x^2 \otimes (I_t I_t^\top)_{\mathrm{circ}}}_{P_{\mathrm{circ}} \text{ (exact 2-D DCT)}} + \underbrace{\frac{\varepsilon}{\Delta t} L_x \otimes \left(e_1 e_1^\top - e_{N_t} e_{N_t}^\top\right)}_{E} - \underbrace{\frac{\varepsilon^2}{4} L_x^2 \otimes \left(e_1 e_1^\top + e_{N_t} e_{N_t}^\top\right)}_{E_2}.$$

Woodbury can handle both E and E_2

In DCT-x mode k, the combined correction is

$$F_k = \left(-\frac{\varepsilon \lambda_x}{\Delta t} - \frac{\varepsilon^2 \lambda_x^2}{4}\right) e_1 e_1^\top + \left(\frac{\varepsilon \lambda_x}{\Delta t} - \frac{\varepsilon^2 \lambda_x^2}{4}\right) e_{N_t} e_{N_t}^\top.$$

Both E and E_2 involve only e_1, e_{N_t}, so F_k is still **rank-2 in time** — the same 2×2 Woodbury system, extended coefficients. The extended Woodbury is then *exactly* correct.

Why Every Naive Discretisation Fails

Even with a correct staggered grid, the first (ad-hoc) projection ignored the rank-2 correction E: it solved $P_{\mathrm{DCT}}\,\phi = f$ via a plain 2-D DCT.

Root cause: the rank-2 boundary term E is dropped

$AA^\top = P_{\mathrm{DCT}} + E$ but the solver treats it as P_{DCT}, leaving a systematic residual

$$\|AA^\top\phi - f\| \;\approx\; \|E\phi\| \;=\; \frac{\varepsilon}{\Delta t}\|L_x\| \cdot |\phi(1) - \phi(N_t)|.$$

For $\varepsilon = 10^{-4}$, $\Delta t = 1/128$: $\varepsilon/\Delta t = 0.013$ — already noticeable. The FP constraint is not satisfied at any grid cell. For $\varepsilon \gtrsim 10^{-4}$ ADMM *diverges*.

Why does ADMM stagnate / diverge?

The returned (ρ, m) satisfies a *wrong* normal equation. The constraint residual stalls at $O(\varepsilon/\Delta t)$ regardless of iteration count. Larger ε makes the violation proportionally worse.

Fix: include *all* boundary corrections (E and E_2) in the solve — either via the extended Woodbury or the banded solver.

Woodbury: Correcting the Rank-2 Boundary Terms

Setup. After DCT-x, mode k decouples. We must solve

$$(P_{\mathrm{circ},k} + F_k)\phi_k = f_k, \quad F_k = \alpha_k\, e_1 e_1^\top + \beta_k\, e_{N_t} e_{N_t}^\top,$$

where $P_{\mathrm{circ},k}$ is *diagonal in DCT-t* and F_k is **rank-2**. The coefficients combining both boundary corrections E and E_2 are:

$$\alpha_k = -\frac{\varepsilon\lambda_x(k)}{\Delta t} - \frac{\varepsilon^2\lambda_x(k)^2}{4}, \qquad \beta_k = \frac{\varepsilon\lambda_x(k)}{\Delta t} - \frac{\varepsilon^2\lambda_x(k)^2}{4}.$$

Woodbury identity. With $U = [e_1, e_{N_t}]$ and $C = \mathrm{diag}(\alpha_k, \beta_k)$:

$$(P_{\mathrm{circ},k} + F_k)^{-1} f_k = \underbrace{P_{\mathrm{circ},k}^{-1} f_k}_{g_k} - P_{\mathrm{circ},k}^{-1}\, U\, M_k^{-1}\, U^\top g_k,$$

where $M_k = C^{-1} + U^\top P_{\mathrm{circ},k}^{-1} U$ is a **precomputed** 2×2 **matrix**.

Algorithm per mode k:

1. $g_k = P_{\mathrm{circ},k}^{-1} f_k$: DCT-$t$, divide by P_{circ} eigenvalues, IDCT-t.
2. Extract boundary values $c_k = [g_k(1);\, g_k(N_t)]$.
3. Solve 2×2 system: $[a_k;\, b_k] = M_k^{-1} c_k$.
4. $\phi_k = g_k - a_k\, d_{1,k} - b_k\, d_{N_t,k}$, where $d_{j,k} = P_{\mathrm{circ},k}^{-1} e_j$ (precomputed).

Steps 1–4 cost $O(N_t \log N_t)$ per mode $\Rightarrow$ $O(N_x N_t \log N_t)$ total.

Woodbury Stability: 2×2 Condition Number

The dominant correction coefficient grows with ε and frequency:

$$|\alpha_k|, |\beta_k| \sim \frac{\varepsilon\, \lambda_x(k)}{\Delta t} \approx \frac{\varepsilon\, k^2}{\Delta x^2\, \Delta t}.$$

For high modes $k \sim N_x$:

$$|\alpha_k| \sim \frac{\varepsilon\, N_x^2\, N_t}{1}.$$

Condition number of M_k: $\kappa(M_k) = O(|\alpha_k|^2)$.
This amplifies rounding errors in $[a_k; b_k]$, corrupting ϕ for high spatial modes.

$N_x = 128,\ \Delta t = 1/128,\ \varepsilon = 0.1$

$|c_{N_x}| \approx 0.1 \times 128^2 \times 128 \approx 2 \times 10^5$
$\kappa(M_{N_x}) \sim 4 \times 10^{10}$

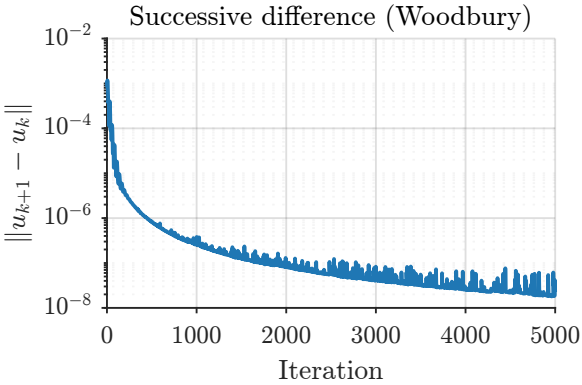
IEEE double: 16 significant digits.
$\Rightarrow \phi$ accurate to ~ 6 digits.
Projection residual $\sim 10^{-6}$, not $\sim 10^{-14}$.

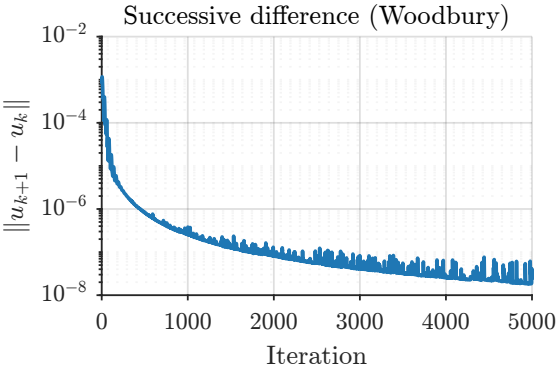
Summary

Woodbury is exact in exact arithmetic; in floating point, the 2×2 condition number grows as $O(\varepsilon^2 N_x^4 N_t^2)$.

Woodbury Results: $N_x = N_t = 128$, $\varepsilon = 10^{-2}$, $\gamma = 1$

Convergence (successive difference $\|u_{k+1} - u_k\|$):





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Note

Woodbury corrects E only (P_{circ} approximates $I_t I_t^\top$). Works at $\varepsilon = 10^{-2}$; degrades for larger ε as $\kappa(M_k)$ grows.

Banded Solver: Block-Diagonalise Only in x

Apply DCT only in x (diagonalise $L_x \to -\lambda_x(k)$). The Kronecker structure $I_{N_x} \otimes (\cdot)$ vs $L_x \otimes (\cdot)$ decouples into N_x independent $N_t \times N_t$ systems.

Mode-k operator (obtained by substituting $L_x \to -\lambda_x(k)$ in A_ρ):

$$A_{\rho,k} = D_t - \varepsilon\, \lambda_x(k)\, I_t, \qquad (A_m A_m^\top)_k = \lambda_x(k)\, I_{N_t}.$$

Per-mode normal equation:

$$T_k\, \hat{\phi}_k = \hat{f}_k, \qquad \boxed{T_k = A_{\rho,k} A_{\rho,k}^\top + \lambda_x(k)\, I_{N_t}}.$$

Expanding $A_{\rho,k} A_{\rho,k}^\top$ and using $D_t I_t^\top + I_t D_t^\top = \frac{1}{\Delta t}(e_1 e_1^\top - e_{N_t} e_{N_t}^\top)$:

$$T_k = D_t D_t^\top - \varepsilon \lambda_x(k) \underbrace{\left(D_t I_t^\top + I_t D_t^\top\right)}_{= \frac{1}{\Delta t}(e_1 e_1^\top - e_{N_t} e_{N_t}^\top)} + \varepsilon^2 \lambda_x(k)^2\, I_t I_t^\top + \lambda_x(k)\, I$$

$$= \underbrace{D_t D_t^\top}_{M_0} + \lambda_x(k) \underbrace{\mathrm{diag}\left(1 - \tfrac{\varepsilon}{\Delta t}, 1, \ldots, 1, 1 + \tfrac{\varepsilon}{\Delta t}\right)}_{M_1} + \lambda_x(k)^2\, \underbrace{\varepsilon^2 I_t I_t^\top}_{M_2}.$$

Banded Solver: Algorithm, DC Mode, and Stability

M_0, M_1, M_2 are all tridiagonal or diagonal $\Rightarrow$ T_k is tridiagonal. Solved in $O(N_t)$ per mode.

Per-call algorithm:

1. DCT in x of $f = A(\mu, \psi)$: cost $O(N_x N_t \log N_x)$.
2. For $k \geq 2$: tridiagonal solve $\hat{\phi}_k = T_k^{-1} \hat{f}_k$. Cost $O(N_x N_t)$ total.
3. **Special case** $k = 1$ ($\lambda_x(1) = 0$, so $T_1 = M_0 = D_t D_t^\top$, singular):

$$\hat{\phi}_1(l) = \begin{cases} 0 & l = 1 \text{ (gauge fix)}, \\ \hat{f}_1(l) \,/\, \lambda_t(l) & l \geq 2. \end{cases} \qquad \text{(DCT-}t\text{ solve, cost } O(N_t \log N_t))$$

4. IDCT in x; apply $A^\top = [I \otimes D_t^\top + \varepsilon L_x \otimes I_t^\top;\ D_x^\top \otimes I]$.

Bug if $k = 1$ is skipped

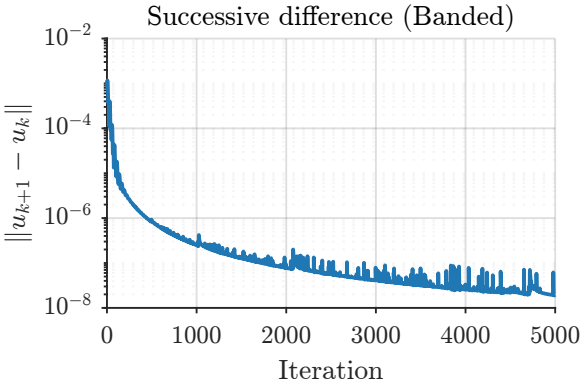
ADMM iterates have non-constant total mass; $\hat{f}_1$ has nonzero non-DC-in-t components. Setting $\hat{\phi}_1 = 0$ leaves these permanently, causing stagnation for all ε.

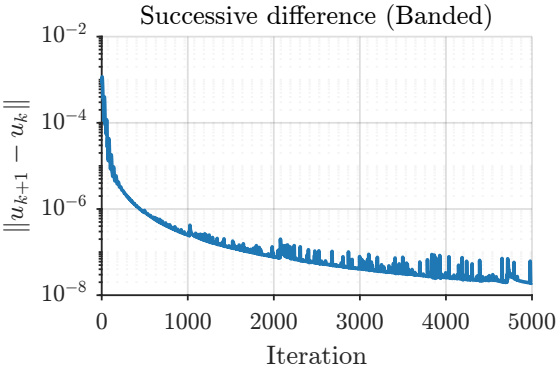
Stability: $\kappa(T_k) = O(N_t^2/\lambda_x(k))$, independent of ε

Since $T_k = A_{\rho,k} A_{\rho,k}^\top + \lambda_x(k) I \succeq \lambda_x(k) I$, the minimum eigenvalue is $\lambda_x(k) > 0$ regardless of ε.

Banded Results: $N_x = N_t = 128$, $\varepsilon = 10^{-2}$, $\gamma = 1$

Convergence (successive difference $\|u_{k+1} - u_k\|$):





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Stability

$\kappa(T_k) = O(N_t^2/\lambda_x(k))$, *independent of* ε. Exact solve — no floating-point amplification.

PCG: Exact Matvec, Approximate Preconditioner

`sb1d_admm_pcg.m`: preconditioner $M{=}P_{\text{circ}}$ (ignores E, E_2); exact matvec $AA^\top\phi$; 30 inner CG iterations per ADMM step.

Why $O(N_x)$ CG iterations are needed

$E = \frac{\varepsilon}{\Delta t} L_x \otimes (e_1 e_1^\top - e_{N_t} e_{N_t}^\top)$ has **rank** $2(N_x{-}1)$ in the full space ($\text{rank}(L_x){=}N_x{-}1$, $\text{rank}(e_1 e_1^\top - e_{N_t} e_{N_t}^\top){=}2$). $P_{\text{circ}}^{-1} AA^\top$ has $O(N_x)$ eigenvalues off from 1; CG needs $O(N_x)$ extra iterations to deflate them.

$N_x{=}128$, 30-iteration cap: only $\sim$23% of outliers handled. CG *stalls*; projection is inaccurate; ADMM diverges.

Outlier eigenvalue

At the DC time mode ($\lambda_t{=}0$), mode k:

$$\mu_{1,k} \approx 1 - \frac{\varepsilon/\Delta t}{1 + \varepsilon^2 \lambda_x(k)}.$$

Largest at $k{=}2$: $\approx 1{-}\varepsilon/\Delta t$.

For $\varepsilon > \Delta t$: $\mu_{1,k} < 0$, system is *indefinite*. CG triggers `pAp`≤ 0 failure and exits with bad ϕ.

Banded solver wins

Exact, $O(N_t)$ per mode, no iteration count tuning.

Four Projection Methods: Side-by-Side

	Naive	**Woodbury**	**PCG**	**Banded**
Exact?	No	Yes	If converged	Yes
Handles E, E_2?	No	Yes	Via matvec	Yes
Large-ε behaviour	Diverges	$\kappa \sim \varepsilon^2 N_x^4 N_t^2$	Stalls/$-$PD	Stable
Failure mode	Missing E	Ill-cond. 2×2	$O(N_x)$ iters needed	None
Cost/call	$O(N^2 \log N)$	$O(N^2 \log N)$	$O(k_{\mathrm{CG}} N^2)$	$O(N^2 \log N_x)$
Precomp.	$O(N^2)$	$O(N^2 \log N_t)$	$O(N^2)$	$O(N^2)$

Recommendation: **banded solver** for all ε (`opts.proj_method = 'banded'`).

Key Takeaways

1. **Optimise first.** Deriving the continuous projection formula $AA^*\phi = f$ (a 4th-order PDE) first, then discretising, reveals the correct adjoint structure. A GPU-friendly discretise-first approach could not be made to work for the SB diffusion coupling.
2. **Dynamic OT avoids Sinkhorn's $e^{-c/\varepsilon}$ blow-up.** Same ε; no exponentially small numbers in the dynamic formulation.
3. **Kronecker structure** $A_\rho = I \otimes D_t + \varepsilon L_x \otimes I_t$, $A_m = D_x \otimes I$ makes the algebra transparent and immediately reveals the $P_{\mathrm{circ}} + E + E_2$ split.
4. **Two rank-2 boundary terms** (E and E_2) arise from the staggered grid. Ignoring either causes ADMM to stall at $O(\varepsilon/\Delta t)$ constraint violation regardless of iteration count.
5. **Extended Woodbury** corrects $E+E_2$ exactly via α_k, β_k, but the 2×2 condition number grows as $O(\varepsilon^2 N_x^4 N_t^2)$ — unstable for large ε.
6. **PCG** uses an exact matvec but an approximate preconditioner. The un-preconditioned correction E has rank $O(N_x)$ in the full space, requiring $O(N_x)$ CG iterations. With a 30-iteration cap, PCG stalls for $\varepsilon/\Delta t \not\ll 1$; for $\varepsilon > \Delta t$ the preconditioned system is indefinite and CG loses positive definiteness.
7. **Banded solver**: exact, $O(N_x N_t \log N_x)$, ε-stable. Strictly superior to all alternatives.